

WORKED EXAMPLE

A neuron can make a Boolean decision

A threshold gate returns 0 or 1 directly; it does not return a probability.

$$s = \sum_i w_i x_i - T = \sum_i w_i x_i + b, \quad b = -T$$

x values are inputs; w values are weights; T is a threshold; b is the equivalent bias; s is the score before the decision.

$$H(s) = 1 \text{ if } s \geq 0, \quad H(s) = 0 \text{ if } s < 0$$

We choose $H(0) = 1$. This inclusive rule reproduces the integer thresholds shown inside the circles on Lecture 2, pages 54-55.

The generic wording "positive" on page 40 can suggest a strict rule. If using a strict rule, shift the thresholds; do not silently mix conventions.

Here we construct fixed weights by hand. Learning those weights is a later topic.

WORKED EXAMPLE

AND: both inputs must be 1

Lecture 2, page 54: weights (1, 1), threshold $T = 2$, bias $b = -2$.

$$y = H(1x_1 + 1x_2 - 2)$$

$$(0, 0) : s = 1(0) + 1(0) - 2 = -2 \Rightarrow H(-2) = 0$$

$$(0, 1) : s = 1(0) + 1(1) - 2 = -1 \Rightarrow H(-1) = 0$$

$$(1, 0) : s = 1(1) + 1(0) - 2 = -1 \Rightarrow H(-1) = 0$$

$$(1, 1) : s = 1(1) + 1(1) - 2 = 0 \Rightarrow H(0) = 1$$

Only the final row reaches the threshold. This is the complete two-input AND truth table.

WORKED EXAMPLE

OR: at least one input must be 1

Lecture 2, page 54: weights (1, 1), threshold $T = 1$, bias $b = -1$.

$$y = H(1x_1 + 1x_2 - 1)$$

$$(0, 0): s = 1(0) + 1(0) - 1 = -1 \Rightarrow H(-1) = 0$$

$$(0, 1): s = 1(0) + 1(1) - 1 = 0 \Rightarrow H(0) = 1$$

$$(1, 0): s = 1(1) + 1(0) - 1 = 0 \Rightarrow H(0) = 1$$

$$(1, 1): s = 1(1) + 1(1) - 1 = 1 \Rightarrow H(1) = 1$$

Changing only the threshold changed AND into OR. The weights still count the number of active inputs.

WORKED EXAMPLE

NOT: reverse a single Boolean input

Lecture 2, page 54: weight -1, threshold $T = 0$, bias $b = 0$.

$$y = H(-x)$$

$$x = 0 : \quad s = (-1)(0) - 0 = 0 \Rightarrow H(0) = 1$$

$$x = 1 : \quad s = (-1)(1) - 0 = -1 \Rightarrow H(-1) = 0$$

A negative weight lets an input inhibit the neuron: turning the input on lowers the score.

NOT means 0 becomes 1 and 1 becomes 0. The thresholded output is a Boolean value, not the signed score.

WORKED EXAMPLE

Majority of three: first four rows

Lecture 2, page 55: at least K inputs active. Here $K = 2$ and all weights are 1.

$$y = H(x_1 + x_2 + x_3 - 2), \quad b = -2$$

$$(0, 0, 0) : s = 0 + 0 + 0 - 2 = -2 \Rightarrow 0$$

$$(0, 0, 1) : s = 0 + 0 + 1 - 2 = -1 \Rightarrow 0$$

$$(0, 1, 0) : s = 0 + 1 + 0 - 2 = -1 \Rightarrow 0$$

$$(0, 1, 1) : s = 0 + 1 + 1 - 2 = 0 \Rightarrow 1$$

The final arrow applies H. The first three rows have fewer than two ones; the fourth row has exactly two.

WORKED EXAMPLE

Majority of three: remaining four rows

Together with the previous page, these cover all $2^3 = 8$ possible inputs.

$$(1, 0, 0): s = 1 + 0 + 0 - 2 = -1 \Rightarrow 0$$

$$(1, 0, 1): s = 1 + 0 + 1 - 2 = 0 \Rightarrow 1$$

$$(1, 1, 0): s = 1 + 1 + 0 - 2 = 0 \Rightarrow 1$$

$$(1, 1, 1): s = 1 + 1 + 1 - 2 = 1 \Rightarrow 1$$

With unit weights, only the number of ones matters, not their positions.

For Boolean inputs, $T = 1.5$ would give the same majority table while avoiding ties. We used $T = 2$ to match the lecture's integer-threshold notation.

INDEPENDENT PRACTICE

Your turn: NAND and a larger vote

NAND returns 0 only when both inputs are 1.

$$(w_1, w_2) = (-1, -1), \quad b = 1.5, \quad T = -1.5$$

1. Calculate the score and output for all four pairs 00, 01, 10, 11. Verify that this is NAND.
2. Construct a unit-weight neuron that returns 1 if at least three out of four inputs are 1. State its threshold and bias.
3. Check the larger gate for input counts 2, 3 and 4.

Remember: $b = -T$, even if the threshold itself is negative.

WORKED ANSWER

Practice answer: check the signs

NAND uses a positive bias and negative input weights.

$$00 : s = -0 - 0 + 1.5 = 1.5 \Rightarrow 1, \quad 01 : s = -0 - 1 + 1.5 = 0.5 \Rightarrow 1$$

$$10 : s = -1 - 0 + 1.5 = 0.5 \Rightarrow 1, \quad 11 : s = -1 - 1 + 1.5 = -0.5 \Rightarrow 0$$

$$y = H(x_1 + x_2 + x_3 + x_4 - 3), \quad T = 3, \quad b = -3$$

$$\text{count} = 2 : s = 2 - 3 = -1 \Rightarrow 0$$

$$\text{count} = 3 : s = 3 - 3 = 0 \Rightarrow 1, \quad \text{count} = 4 : s = 4 - 3 = 1 \Rightarrow 1$$

Avoid: using T as the bias without changing its sign; treating score 0 as output 0 under our inclusive convention.